

High LDL cholesterol confers greater risk of ST-segment elevation myocardial infarction than non-ST-segment elevation myocardial infarction in statin-treated patients with ischaemic heart disease

Malene Kærslund Hansen ^{1,2*}, Martin Bødtker Mortensen^{1,3},
Kevin Kris Warnakula Olesen¹, Pernille Gro Thrane ^{1,2},
Malene Højgaard Andersen^{1,2}, Christine Gyldenkerne ^{1,2},
Nina Stødkilde-Jørgensen ^{1,2}, and Michael Maeng ^{1,2}

¹Department of Cardiology, Aarhus University Hospital, Palle Juul-Jensens Boulevard 99, Aarhus N 8200, Denmark; ²Department of Clinical Medicine, Health, Aarhus University, Palle Juul-Jensens Boulevard, Aarhus N 8200, Denmark; and ³Johns Hopkins Ciccarone Center for the Prevention of Cardiovascular Disease, Johns Hopkins University School of Medicine, Baltimore, MD, USA

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Aims

LDL cholesterol (LDL-C) is causally associated with myocardial infarction (MI). However, MI covers two clinically different entities: ST-segment elevation MI (STEMI) and non-STEMI (NSTEMI). We aimed to assess whether high LDL-C confers greater risk of STEMI than NSTEMI in statin-treated patients with ischaemic heart disease (IHD).

Methods and results

We included statin-treated patients with IHD determined by coronary angiography from the Western Denmark Heart Registry between 2011 and 2020. LDL-C and non-HDL cholesterol (non-HDL-C) levels were measured within 1 year after coronary angiography. The risk of STEMI and NSTEMI was estimated as adjusted hazard ratios (aHRs). The comparison of STEMI vs. NSTEMI and 30 day mortality after STEMI vs. NSTEMI was estimated as adjusted odds ratios (aORs). The study included 36 739 statin-treated patients with IHD: 26 178 (71%) men and median age 66 years. During median follow-up of 4.9 years, 531 STEMI and 1614 NSTEMI events occurred. Per 1 mmol/L higher LDL-C, the aHRs of STEMI and NSTEMI were 1.43 [95% confidence interval (CI): 1.30–1.57] and 1.23 (95% CI: 1.16–1.31), corresponding to an aOR of 1.18 (95% CI: 1.04–1.33) for STEMI vs. NSTEMI. Patients at LDL-C goal ≤ 1.4 mmol/L vs. >2.2 mmol/L had a lower risk of STEMI and NSTEMI, with 22% lower odds of STEMI than NSTEMI. Results were similar for non-HDL-C. STEMI was associated with higher 30 day mortality than NSTEMI: aOR 1.62 (95% CI: 1.02–2.57).

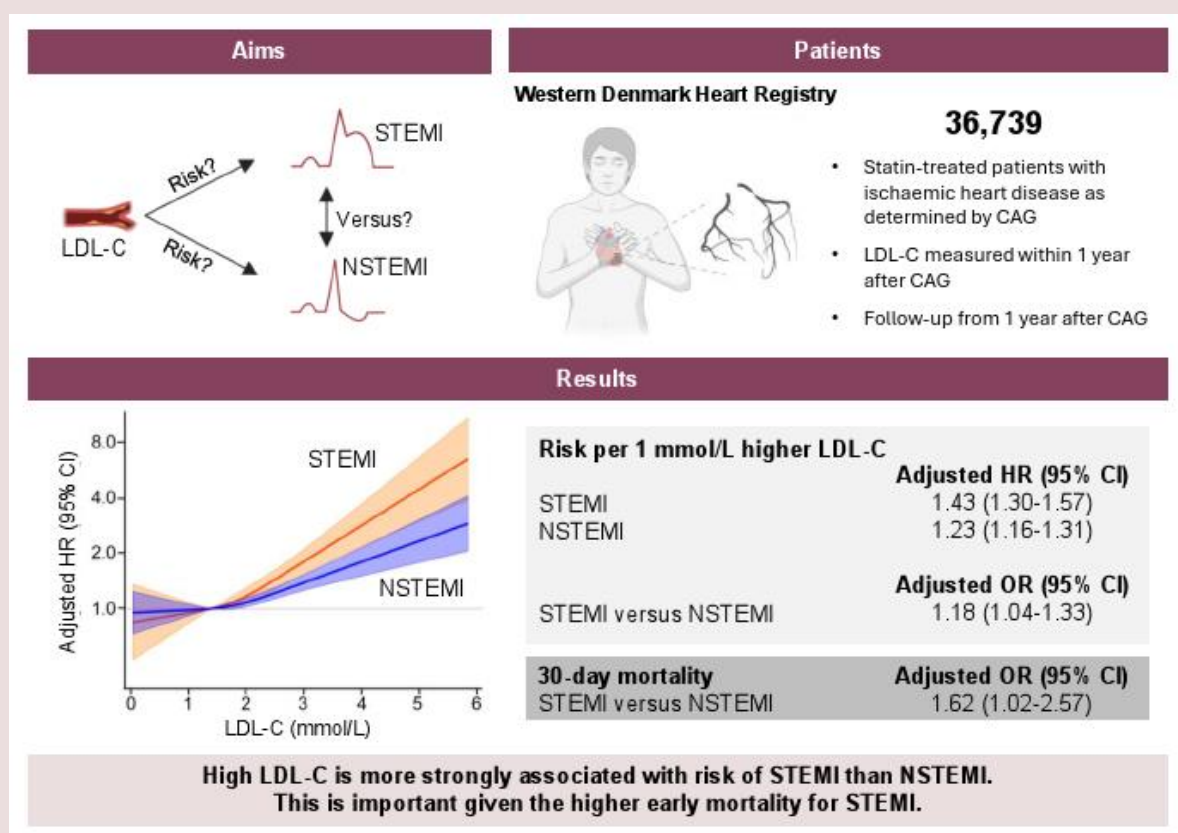
Conclusion

High LDL-C confers greater risk of STEMI than NSTEMI in statin-treated patients with IHD. This is important given the higher early mortality associated with STEMI.

* Corresponding author. Email: makaha@clin.au.dk

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Graphical Abstract



Among 36 739 statin-treated patients with ischaemic heart disease as determined by CAG, a 1 mmol/L higher LDL-C confers greater risk of STEMI (adjusted HR 1.43) and NSTEMI (adjusted HR 1.23), corresponding to 18% higher odds of STEMI than NSTEMI. This is clinically relevant given the higher early mortality associated with STEMI (adjusted OR 1.62) and suggests that cholesterol lowering reduce the risk of STEMI to a greater extent than NSTEMI. CAG, coronary angiography; HR, hazard ratio; OR, odds ratio. Illustrations created with biorender.com.

Keywords

LDL-C • Non-HDL-C • Statin-treated • Ischaemic heart disease • STEMI • NSTEMI

Introduction

LDL cholesterol (LDL-C) levels are causally associated with myocardial infarction, and evidence from numerous randomized controlled trials has proven that statin treatment lowers LDL-C levels and reduces the risk of myocardial infarction.^{1,2} For research purposes, myocardial infarction is considered a single endpoint. However, in clinical practice, we carefully distinguish between the two different types of myocardial infarction—ST-segment elevation myocardial infarction (STEMI) and non-STEMI (NSTEMI)—given their differing prognoses and treatment strategies.

Atherosclerosis is the underlying cause of both STEMI and NSTEMI and is characterized by cholesterol deposition and plaque formation in the coronary arteries.³ In STEMI, the sudden rupture of a cholesterol-rich atherosclerotic plaque often causes thrombus formation and a total coronary artery occlusion.⁴⁻⁶ In contrast, an NSTEMI may be preceded by the rupture or erosion of less cholesterol-rich atherosclerotic plaques, mainly causing non-occlusive coronary artery lesions.^{5,6} This difference explains the higher early mortality associated with STEMI compared with NSTEMI and the indication for acute therapeutic intervention recommended for patients presenting with a STEMI compared with the subacute evaluation recommended for most patients presenting with an NSTEMI.⁷

Statins promote atherosclerotic plaque stabilization and plaque regression, with treatment efficacy assessed by LDL-C and non-HDL

cholesterol (non-HDL-C) levels.^{8,9} The association between LDL-C and non-HDL-C levels with risk of STEMI and NSTEMI in statin-treated patients with ischaemic heart disease (IHD) remains controversial. Leveraging the Western Denmark Heart Registry, we aimed to test the hypothesis that high LDL-C and non-HDL-C levels confer greater risk of STEMI than NSTEMI in statin-treated patients with IHD.

Methods

This cohort study was conducted using data from high-quality national and regional health registries in Denmark. The core registry from which the study population was sampled—the Western Denmark Heart Registry—includes all coronary angiographies performed in Western Denmark since 1999.¹⁰ Pre-specified patient characteristics and clinical procedural information were collected at the time of examination by the treating physician. The unique 10-digit Central Personal Registry number assigned to each Danish resident at birth or immigration allows for accurate individual-level linkage between Danish regional and national health registries. The Danish healthcare system excels by equal access to healthcare for all residents through tax-funded services provided by the Danish National Health Service.

This study was approved by the Danish Data Protection Agency (record no. 1-16-02-193-18). According to Danish regulations, observational registry-based studies do not require approval from the ethics committee and informed consent from participants was exempted (record no: 1-45-70-24-22).

Study population

The study population included all patients aged 18 years and older with symptomatic IHD—STEMI, NSTEMI, unstable angina pectoris, or stable angina pectoris—and confirmed coronary artery disease, as determined by first-time coronary angiography (CAG) between 1 January 2011, and 31 December 2020 (see [Supplementary material online, Figure S1](#)).

Index cholesterol and triglyceride measurements were chosen as the latest available complete lipid panel within 1 year after CAG to obtain stable LDL-C and non-HDL-C levels. The complete lipid panel included total cholesterol, HDL-C, LDL-C, and triglycerides. LDL-C was calculated using the Friedewald Formula (total cholesterol—HDL cholesterol—triglycerides/2.2 in mmol/L) and non-HDL-C as total cholesterol—HDL-C. Remnant cholesterol was calculated as non-HDL-C—LDL-C. Cholesterol levels were measured in SI units and collected from the nationwide Register of Laboratory Results for Research, which holds information on all blood samples from hospitals and general practitioners.¹⁰

Statin treatment was defined as a redeemed prescription of any statin between 30 and 365 days before the index cholesterol date. The 30 day quarantine period ensured a steady state of the drug. Data on medical therapy was collected from the Danish National Prescription Registry, which holds information on all reimbursed prescriptions issued by the Danish National Health Service (see [Supplementary material online, Table S1](#)).

Patients with triglycerides >4.5 mmol/L (for whom the Friedewald formula is not applicable) or those not receiving statin treatment were excluded from the study population.

Study outcomes

STEMI and NSTEMI were defined using selected subsections of the International Classification of Disease (ICD)—10th edition code I21 registered as a primary discharge diagnosis from an acute hospital admission in combination with the surgical procedure codes for primary percutaneous coronary intervention (PCI) (see [Supplementary material online, Table S1](#)). Diagnoses were obtained from the Danish National Patient Register; a national registry with complete information on all hospital discharge codes since 1977.¹⁰ Additionally, 30 day mortality after STEMI and NSTEMI was estimated. Information on all-cause death and emigration was obtained from the Civil Registration System.¹¹

Covariate definitions are included in [Supplementary methods](#).

Statistical analyses

Follow-up began 1 year after CAG and continued until STEMI, NSTEMI, death, emigration ($n = 94$), or end of follow-up (18 April 2023), whichever came first. Baseline characteristics were presented as numbers and proportions for categorical variables and as medians with first and third quartiles for continuous variables. To test the study hypothesis, we employed two different designs: (i) a population-based cohort design to assess the risk of outcomes and (ii) a case-control design to compare the strength of the association between LDL-C and non-HDL-C levels with STEMI vs. NSTEMI. The corresponding statistical methods were as follows: (i) cause-specific Cox proportional hazards regression analyses examining the risk of STEMI and NSTEMI and (ii) logistic regression analyses examining the association between LDL-C and non-HDL-C levels with STEMI vs. NSTEMI and the 30 day mortality after STEMI vs. NSTEMI.¹²

For the Cox proportional hazard regression models, time since start of follow-up was used as the underlying time scale. The proportional hazards assumption was examined graphically using log-log plots for categorical variables and by including a time-dependent interaction term for continuous variables. No violations were observed. Hazard ratios (HRs) for STEMI and NSTEMI were presented in restricted cubic spline models by LDL-C and non-HDL-C levels with three knots at the 10th, 50th, and 90th percentile (chosen using the Bayesian information criterion to balance best fit and complexity,^{13,14}) per 1 mmol/L higher LDL-C and non-HDL-C, by LDL-C and non-HDL-C percentile cut-off (<25th, 25–74th, 75–94th, and ≥95th), and by secondary prevention cholesterol goal achievement (yes vs. no; LDL-C ≤ 1.4 mmol/L vs. >2.2 mmol/L and non-HDL-C ≤ 2.2 mmol/L vs. >3.0 mmol/L), respectively.

Among patients who had a myocardial infarction (the first event of either a STEMI or NSTEMI) during follow-up, the odds ratio (OR) of STEMI vs. NSTEMI was estimated per 1 mmol/L higher LDL-C and non-HDL-C, in restricted cubic spline models, and by secondary prevention cholesterol goal achievement.

The pre-specified multivariable adjustment for both the Cox regression models and logistic regression models included age (<50, 50–59, 60–69,

70–79, and >80 years), sex, smoking (active, former, and never), hypertension, and diabetes. Adjustment of LDL-C additionally included remnant cholesterol. Missing values for covariates in the adjustment models (smoking 8.9%) were handled by multiple imputations using chained equations, generating nine imputations. Missing values were assumed to be missing at random, and imputation was based on the variables listed in [Supplementary material online, Table S2](#).

Event rates per 1000 person-years were reported as crude risk estimates of STEMI and NSTEMI in the overall cohort and by LDL-C and non-HDL-C goal achievement.

As sensitivity analyses, we stratified by baseline characteristics: sex (men/women), age (<65/≥65 years), diabetes, hypertension, indication for CAG (STEMI/NSTEMI or unstable angina pectoris/stable angina pectoris), revascularization within 90 days after CAG, and year of CAG (2011–15/2016–20). Index LDL-C and non-HDL-C levels by year of CAG and the proportion of patients receiving high-intensity statins by year of CAG are reported. A complete case analysis was performed, and ezetimibe users were compared with non-ezetimibe users. Analyses were conducted using STATA/MP version 18.0 (StataCorp LP, College Station, TX, USA) and graphical illustrations were created in GraphPad Prism Version 10.4.1.

Results

Based on a study population of 55 878 patients examined by CAG due to IHD between 1 January 2011, and 31 December 2020, we included 36 739 statin-treated patients with complete lipid panels in the study cohort (see [Supplementary material online, Figure S2](#)). Of these, 26 178 (71%) were men and the median age was 66 years (Q1–Q3: 58–74) ([Table 1](#)). The indication for CAG was acute coronary syndrome in 21 028 (57%) patients, 26 196 (71%) patients received high-intensity statins, and 11 247 patients (31%) reached the guideline-recommended LDL-C goal within 1 year after CAG.

During a median follow-up of 4.9 years (Q1–Q3: 3.0–7.0), STEMI was observed in 531 patients and NSTEMI in 1614 patients (see [Supplementary material online, Figure S3](#)). Event rates per 1000 person-years of STEMI and NSTEMI were 2.8 (95% CI: 2.6–3.1) and 8.7 (95% CI: 8.3–9.2), respectively.

High LDL cholesterol and risk of ST-segment elevation myocardial infarction and non-ST-segment elevation myocardial infarction

In the spline model, a higher LDL-C level was associated with an increased risk of both STEMI and NSTEMI, with the risk of STEMI exceeding that of NSTEMI ([Figure 1](#)). Per 1 mmol/L higher LDL-C, the adjusted HRs of STEMI and NSTEMI were 1.43 (95% CI: 1.30–1.57) and 1.23 (95% CI: 1.16–1.31), respectively ([Figure 2](#); see [Supplementary material online, Table S3](#)). A higher LDL-C percentile was associated with an increased risk of both STEMI and NSTEMI, with the risk of STEMI exceeding that of NSTEMI at LDL-C levels >75th percentile (see [Supplementary material online, Figure S4](#)). Results were consistent in subgroup analyses stratified by sex, age, diabetes, hypertension, acute coronary syndrome and stable angina, revascularization within 90 days after CAG, and year of CAG (see [Supplementary material online, Figure S5](#) and [Table S4](#)).

High LDL cholesterol and risk of ST-segment elevation myocardial infarction vs. non-ST-segment elevation myocardial infarction

A higher LDL-C level was more strongly associated with STEMI than NSTEMI: Per 1 mmol/L higher LDL-C, the adjusted OR of STEMI vs. NSTEMI was 1.18 (95% CI: 1.04–1.33) ([Figure 3](#); see [Supplementary material online, Table S5](#)). In the spline model, an LDL-C level above ~2 mmol/L was associated with higher odds of STEMI than NSTEMI, while there were no differences in odds below this level

Table 1 Baseline characteristics of statin-treated patients with ischaemic heart disease from the Western Denmark Heart Registry

	Total study cohort	LDL-C at goal		Non-HDL-C at goal	
		Yes ≤ 1.4 mmol/L	No > 2.2 mmol/L	Yes ≤ 2.2 mmol/L	No > 3.0 mmol/L
Patients	36 739 (100)	11 247 (100)	7743 (100)	15 179 (100)	6534 (100)
Men	26 178 (71.3)	8701 (77.4)	4941 (63.8)	11 548 (76.1)	4191 (64.1)
Women	10 561 (28.7)	2546 (22.6)	2802 (36.2)	3631 (23.9)	2343 (35.9)
Age, years	66 (58–74)	66 (58–74)	66 (58–74)	67 (59–74)	66 (57–73)
BMI, kg/m ^{2,a}	27 (24–30)	27 (25–30)	27 (24–30)	27 (24–29)	27 (25–31)
Diabetes	6988 (19.0)	2696 (24.0)	1238 (16.0)	2847 (18.8)	1316 (20.1)
Hypertension	22 083 (60.1)	6655 (59.2)	4800 (62.0)	8566 (56.4)	4207 (64.4)
Family history of coronary heart disease ^a	14 185 (38.6)	4033 (35.9)	3225 (41.7)	5398 (35.6)	2755 (42.2)
Smoking ^a					
Active	9913 (27.0)	3044 (27.1)	2217 (28.6)	3858 (25.4)	2022 (30.9)
Former	13 600 (37.0)	4086 (36.3)	2848 (36.8)	5559 (36.6)	2358 (36.1)
Never	9947 (27.1)	3092 (27.5)	2044 (26.4)	4384 (28.9)	1599 (24.5)
Indication for CAG					
STEMI	9271 (25.2)	3374 (30.0)	1514 (19.6)	4451 (29.3)	1288 (19.7)
NSTEMI + unstable angina pectoris	11 757 (32.0)	3720 (33.1)	2389 (30.9)	4940 (32.5)	2028 (31.0)
Stable angina pectoris	15 711 (42.8)	4153 (36.9)	3840 (49.6)	5877 (38.1)	3218 (49.3)
Revascularization ≤ 90 days after CAG	27 572 (75.0)	8938 (79.5)	5175 (66.8)	12 018 (79.2)	4400 (67.3)
PCI	23 762 (64.7)	7859 (69.9)	4396 (56.8)	10 530 (69.4)	3761 (57.6)
CABG	4571 (12.4)	1298 (11.5)	953 (12.3)	1778 (11.7)	779 (11.9)
Medication					
High-intensity statin ^b	26 196 (71.3)	9075 (80.7)	4600 (59.4)	12 136 (80.0)	3976 (60.9)
Low-intensity statin	10 543 (28.7)	2172 (19.3)	3143 (40.6)	3043 (20.0)	2558 (39.1)
Ezetimibe	2795 (7.6)	954 (8.5)	596 (7.7)	1269 (8.4)	538 (8.2)
Aspirin	32 953 (89.7)	10 235 (91.0)	6785 (87.6)	13 787 (90.8)	5712 (87.4)
Cholesterol measures					
Non-HDL-C, mmol/L	2.3 (1.9–2.8)	1.7 (1.5–2.0)	3.4 (3.0–3.9)	1.8 (1.6–2.0)	3.5 (3.2–4.1)
LDL-C, mmol/L	1.7 (1.3–2.1)	1.2 (1.0–1.3)	2.6 (2.4–3.1)	1.3 (1.1–1.5)	2.7 (2.4–3.2)
Remnant-C, mmol/L	0.6 (0.4–0.8)	0.6 (0.4–0.8)	0.6 (0.5–0.9)	0.5 (0.3–0.6)	0.9 (0.6–1.2)
Triglycerides, mmol/L	1.2 (0.9–1.8)	1.2 (0.9–1.8)	1.4 (1.1–2.0)	1.0 (0.8–1.3)	1.9 (1.4–2.6)

Numbers (%) for categorical variables and median (first and third quartiles) for continuous variables.

BMI, body mass index; CABG, coronary artery bypass grafting; CAG, coronary angiography; LDL-C, LDL cholesterol; non-HDL-C, non-HDL cholesterol; NSTEMI, non-ST-segment elevation myocardial infarction; PCI, percutaneous coronary intervention; remnant-C, remnant cholesterol; STEMI, ST-segment elevation myocardial infarction.

^aMissing data in the total study cohort: BMI $n = 1417$ (3.9%), family history of coronary heart disease $n = 3776$ (10.3%), and smoking $n = 3279$ (8.9%).

^bAtorvastatin 40–80 mg, rosuvastatin 20–40 mg.

(see [Supplementary material online, Figure S6](#)). The magnitude of the association was consistent across various subgroup analyses, although it was attenuated among patients aged 65 years or older and was not observed among patients with diabetes (see [Supplementary material online, Figure S7](#) and [Table S6](#)).

LDL cholesterol goal achievement and risk of ST-segment elevation myocardial infarction vs. non-ST-segment elevation myocardial infarction

Among patients who achieved the guideline-recommended LDL-C goal compared with those not at goal, there was a higher prevalence of men, STEMI at index CAG, and treatment with high-intensity statins

([Table 1](#)). During the study period, the proportion of patients achieving the LDL-C goal increased concomitantly with an increased use of high-intensity statins (see [Supplementary material online, Figures S8](#) and [S9](#)). Patients at goal, compared with those not at goal, had a lower risk of STEMI and NSTEMI; adjusted HRs of 0.53 (95% CI: 0.42–0.67) and 0.69 (95% CI: 0.60–0.79), respectively ([Figure 4](#); see [Supplementary material online, Figure S10](#)). Patients achieving the LDL-C and non-HDL-C goals, compared with those not achieving goal, had 22% and 27% lower odds of STEMI than NSTEMI; adjusted OR 0.78 (95% CI: 0.58–1.04) and 0.73 (95% CI: 0.55–0.95), respectively ([Figure 5](#)).

Patients achieving the LDL-C goal, compared with those not achieving goal, had lower event rates per 1000 person-years of STEMI [2.4 (95% CI: 2.0–2.9) and 4.3 (95% CI: 3.8–5.0)] and NSTEMI [8.0 (95% CI: 7.3–8.8) and 11.4 (95% CI: 10.4–12.4)] ([Figure 6](#); see [Supplementary material online, Table S7](#)), with a relatively greater decline for STEMI

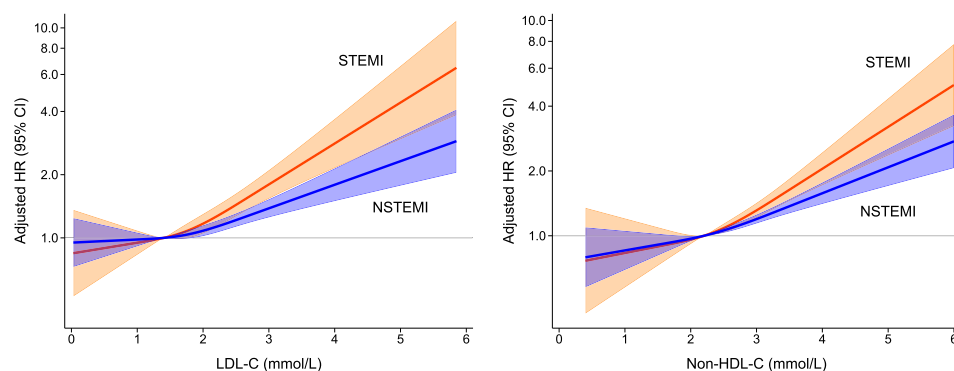


Figure 1 Adjusted hazard ratios (solid lines) and 95% confidence intervals (scattered areas) for ST-segment elevation myocardial infarction and non-ST-segment elevation myocardial infarction according to LDL cholesterol and non-HDL cholesterol levels in statin-treated patients with ischaemic heart disease in restricted cubic spline models. Reference levels were chosen at 1.4 and 2.2 mmol/L for LDL cholesterol and non-HDL cholesterol, respectively. To avoid extremely wide confidence intervals, patients with LDL cholesterol >6.0 mmol/L ($n = 20$) and non-HDL cholesterol <0.5 or >6.0 mmol/L ($n = 98$) were excluded from respective analyses.

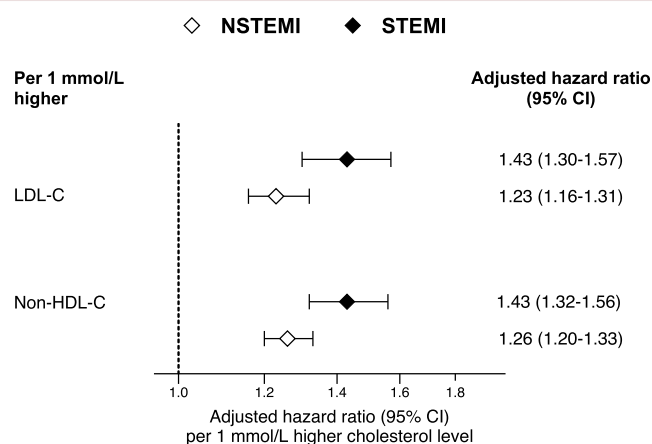


Figure 2 Adjusted hazard ratios for ST-segment elevation myocardial infarction and non-ST-segment elevation myocardial infarction per 1 mmol/L higher LDL cholesterol and non-HDL cholesterol in statin-treated patients with ischaemic heart disease from the Western Denmark Heart Registry. Adjustment included age, sex, smoking, hypertension, diabetes, and for LDL-C, additionally, remnant cholesterol.

than NSTEMI. Results were similar for non-HDL-C and by age group below or above 65 years (see [Supplementary material online, Figure S11](#)).

All results were similar for non-HDL-C and in complete case analyses (see [Supplementary material online, Table S8](#)). Ezetimibe users compared with non-ezetimibe users had lower odds of STEMI than NSTEMI: adjusted OR 0.74 (95% CI: 0.45–1.21) (see [Supplementary material online, Table S9](#)).

Early mortality after ST-segment elevation myocardial infarction vs. non-ST-segment elevation myocardial infarction

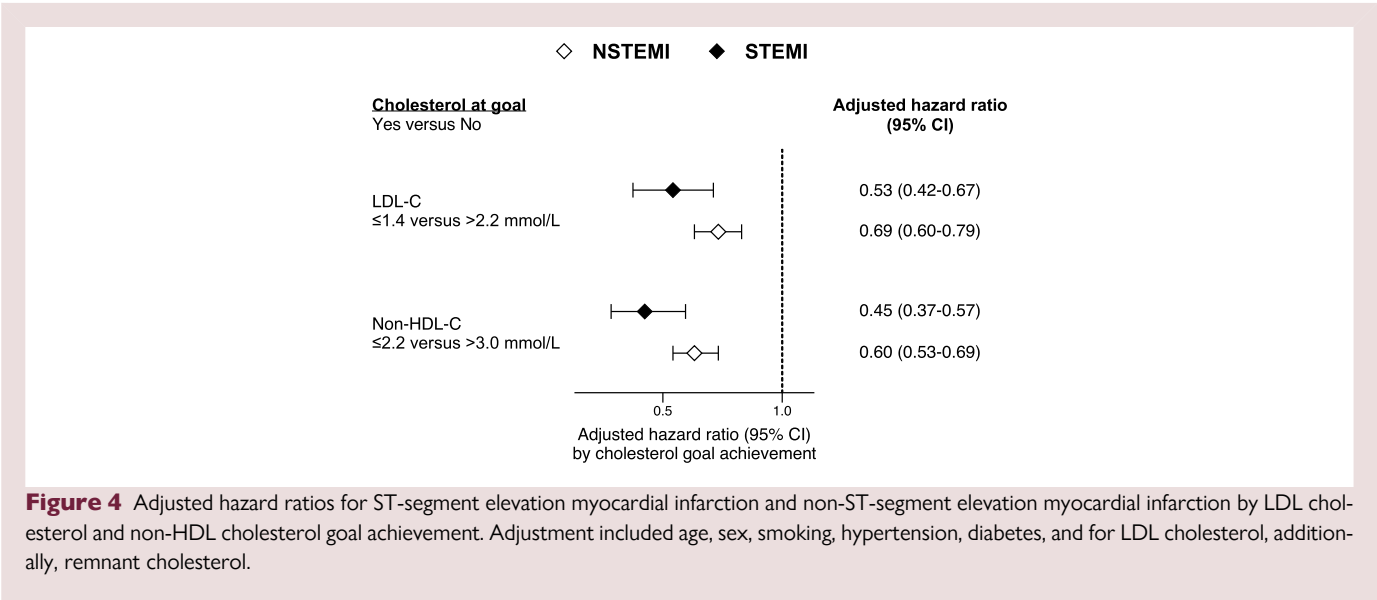
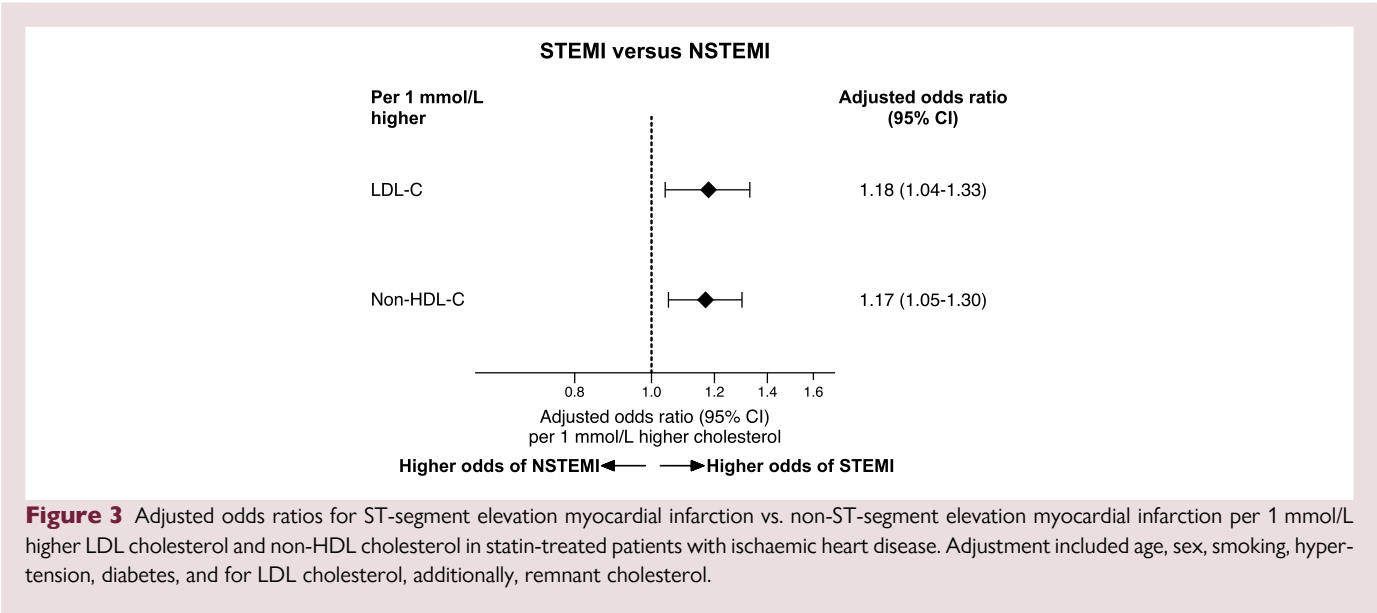
Among statin-treated patients with IHD, those who had a STEMI compared with an NSTEMI during follow-up were younger (median age 68 years vs. 74 years), more frequently men (78% vs. 68%), and more often

active smokers (40% vs. 29%). STEMI was associated with higher 30 day mortality than NSTEMI: adjusted OR 1.62 (95% CI: 1.02–2.57) (see [Supplementary material online, Table S10](#)).

Discussion

In this large contemporary population-based cohort comprising more than 36 000 statin-treated patients with IHD, we provide novel data demonstrating that higher LDL-C and non-HDL-C levels confer greater risk of STEMI than NSTEMI. Specifically, we found that (i) higher LDL-C and non-HDL-C levels increased the risk of STEMI more than NSTEMI, (ii) patients treated to secondary prevention LDL-C and non-HDL-C goals had a greater reduction of STEMI than NSTEMI events, and (iii) STEMI was associated with higher 30 day mortality than NSTEMI.

Our data are based on cross-linkage of high-quality healthcare registries within a well-organized Danish healthcare system that enables us



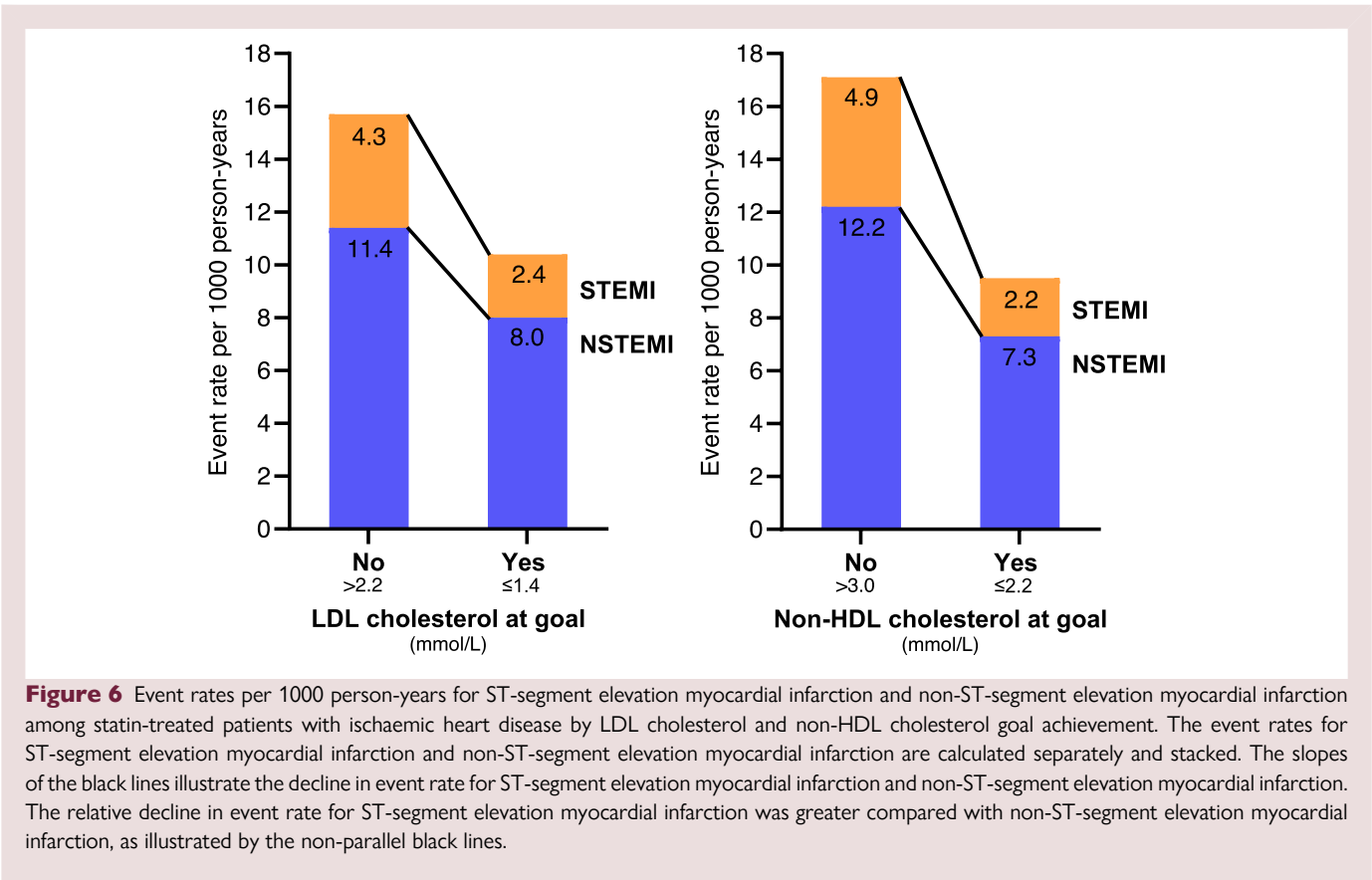
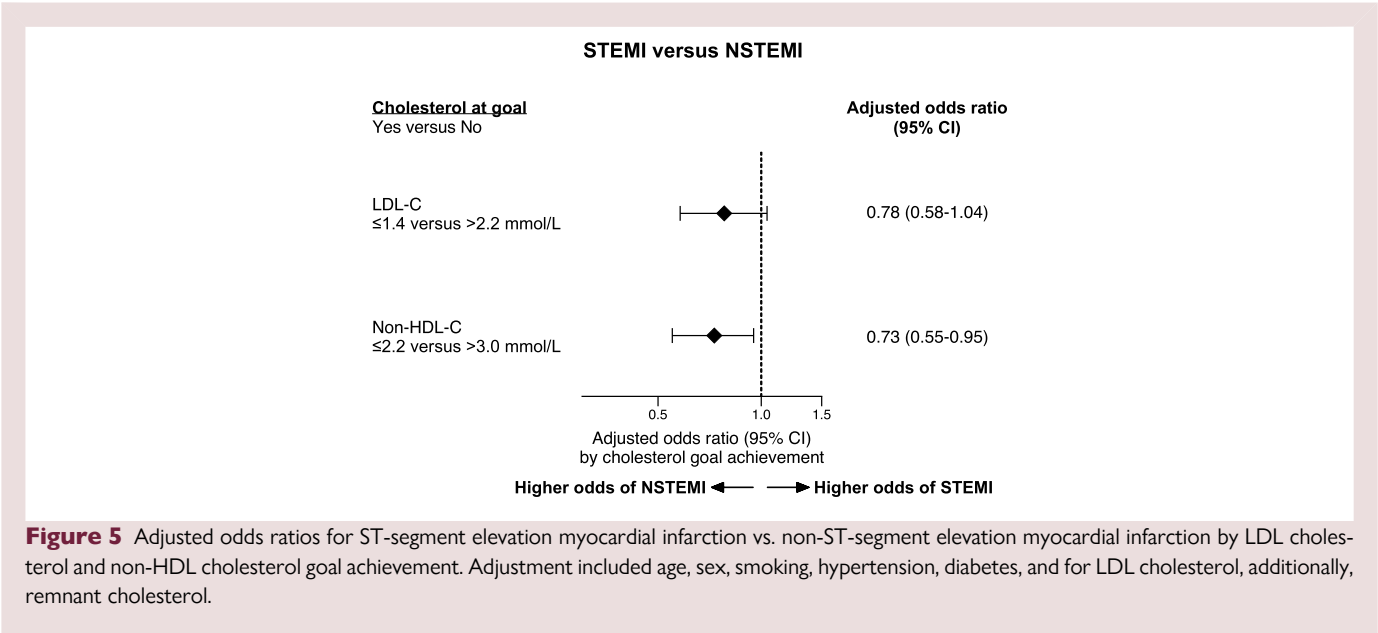
to investigate STEMI and NSTEMI as separate outcomes. Our findings indicate that statin-induced lowering of LDL-C and non-HDL-C levels reduces the risk of STEMI more than NSTEMI and thereby may decrease cardiovascular mortality.

Clinical perspectives

Our study is the first to provide real-world evidence of the association between LDL-C and non-HDL-C levels with the risk of STEMI vs. NSTEMI in secondary prevention. However, recently, a Danish nationwide primary prevention cohort study showed that current statin use, compared with never statin use, was associated with a lower risk of STEMI than NSTEMI [aOR 0.81 (95% CI: 0.77–0.85)].¹⁵ Together with our findings, these two studies provide robust evidence of a shift in the clinical presentation of myocardial infarction from STEMI towards NSTEMI with statin use and lower LDL-C levels in both primary and secondary prevention settings.

Previous randomized controlled trials have documented the proportional relationship between statin-induced lower LDL-C levels and a lower risk of myocardial infarction.^{1,2} This study adds to evidence that the proportional relationship for LDL-C may differ by type of myocardial infarction. Meaning, with higher LDL-C levels the risk of STEMI increases more than that of NSTEMI and, contrarily, with lower LDL-C levels, the risk of STEMI decreases more than that of NSTEMI. Therefore, using myocardial infarction as a single endpoint constitutes a significant simplification, whereas assessing the risk of STEMI and NSTEMI separately provides clinically important insights into the relationship with LDL-C levels.

International guidelines provide well-defined treatment goals for LDL-C and non-HDL-C to facilitate treatment adherence.¹⁶ In 2019, the secondary prevention treatment goal for LDL-C was lowered from ≤1.8 mmol/L (70 mg/dL) to ≤1.4 mmol/L (55 mg/dL) together with the inclusion of a non-HDL-C treatment goal ≤2.2 mmol/L (85 mg/dL) to address cholesterol-related residual risk not captured by LDL-C alone.¹⁶ Already by 2020, ~40% of patients in this cohort



achieved the LDL-C goal ≤ 1.4 mmol/L. Contemporary data from secondary prevention cohorts in the USA and Europe report that 30 and 37% of patients, respectively, achieve this LDL-C goal.^{17,18} These data suggest an urgent need to optimize LDL-C lowering in patients with established cardiovascular disease as a simple and cost-effective way to reduce any myocardial infarction and, in particular, STEMI. Moreover,

recent European data indicate that early statin plus ezetimibe combination therapy both increases the proportion of patients achieving guideline-recommended LDL-C targets and confers a lower risk of future cardiovascular outcomes.^{19,20} Based on the results from this study, physicians must overcome any reluctance to intensify preventive medical therapy, such as

adjusting statin dose, adding ezetimibe, and, when indicated, introducing proprotein convertase subtilisin/kexin type 9 inhibitors to achieve guideline-recommended LDL-C levels. Continuous monitoring of LDL-C and non-HDL-C levels is essential to ensure achievement of these treatment goals. Moreover, these findings add perspective to the dialogue between patients and physicians regarding risk assessment and risk reduction. Our data may serve as a contemporary basis for patient information and education, where the conversation may not only focus on the recurrent risk of any myocardial infarction but also specify the type of myocardial infarction.²¹

Atherosclerosis in ST-segment elevation myocardial infarction and non-ST-segment elevation myocardial infarction

Besides being clinically relevant to cardiac rehabilitation and risk management programmes, these results frame a clinical perspective of the fundamental properties of atherosclerosis. The theory of the 'vulnerable plaque rupture' has traditionally been the concept of mind defining the pathophysiological process behind acute myocardial infarction. This concept is based on autopsy studies of patients with sudden cardiac death in an era where most myocardial infarctions were STEMI events.^{6,22} However, during the past decades, the presentation of myocardial infarction has shifted, with NSTEMI now comprising the majority of events instead of STEMI.⁶ This trend has evolved alongside preventive initiatives like the use of high-intensity statins. Statins may lower LDL-C by 30–60%, leading to plaque regression proportional to the reduction in LDL-C.^{9,16} The statin-induced plaque remodelling and stabilization is provided by a decreased plaque lipid content, increased calcified plaque volume, and fibrous cap thickening.⁹ Altogether, the stabilized plaque promotes a shift towards plaque erosion rather than plaque rupture, incomplete rather than complete coronary artery occlusions, and the clinical presentation of myocardial infarction being NSTEMI rather than STEMI.^{6,23} Our data demonstrate that LDL-C and non-HDL-C levels may be clinical markers of such underlying mechanisms in statin-treated patients with IHD.

Limitations

First, we acknowledge that the observational study design precludes a causal interpretation. Yet, as causality has already been proven between LDL-C levels and atherosclerosis burden, along with evidence showing that intense lipid-lowering therapy reduces atheroma burden,^{1,2,8} we find it plausible that our results reflect causality. Second, our main findings rely on the accuracy of diagnostic coding. While diagnoses of myocardial infarctions have high positive predictive values in Danish registries, the accuracy of diagnostic coding for recurrent STEMI and NSTEMI is unknown.^{24,25} To increase the outcome validity for STEMI, we included both ICD-10 codes and procedure codes of primary PCI, as primary PCI has been the reperfusion strategy for STEMI in Denmark since 2003.²⁶ Third, we chose to evaluate late outcomes following CAG to assess the impact of achieved LDL-C and non-HDL-C levels once statin therapy had exerted its protective effects on coronary plaques. And fourth, contrary to current clinical practice, ezetimibe use was uncommon in Denmark during the study period; hence, the respective analyses may be biased by confounding by indication.

Conclusions

In statin-treated patients with IHD, high LDL-C confers greater risk of STEMI than NSTEMI. Our findings indicate that statin-induced lowering of LDL-C and non-HDL-C levels reduces the risk of STEMI more than

NSTEMI. This is important given that STEMI is associated with higher early mortality than NSTEMI.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

Author contributions

Malene Kærslund Hansen (Conceptualization [supporting]; Formal analysis [lead]; Methodology [lead]; Project administration [lead]; Writing—original draft [lead]), Martin Bødtker Mortensen (Conceptualization [lead]; Methodology [supporting]), Kevin Kris Warnakula Olesen (Data curation [lead]; Formal analysis [supporting]; Methodology [supporting]; Writing—review and editing [supporting]), Pernille Gro Thrane (Methodology [supporting]; Writing—review and editing [supporting]), Malene Højgaard Andersen (Visualization [supporting]; Writing—review and editing [supporting]), Christine Gyldenkerne (Methodology [supporting]; Writing—review and editing [supporting]), Nina Stødkilde-Jørgensen (Writing—review and editing [supporting]), and Michael Maeng (Conceptualization [supporting]; Funding acquisition [lead]; Methodology [supporting]; Writing—review and editing [lead])

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Data availability

Open access to data is not allowed by the Danish Data Protection Agency. However, given a reasonable request, additional analyses can be done after contacting the corresponding author.

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